

STRATEGIC BRIEFING: CAPTAIN ONBOARDING CONVERSION & AIRPORT SUPPLY ARCHITECTURE

TO: Head of Supply, Rapido FROM: Senior Data Scientist, Marketplace Strategy DATE: September 8, 2026 SCOPE: 25,000 Signups & 60,000 Airport Trips

EXECUTIVE TAKEAWAYS

- **The Onboarding Funnel is Bleeding at Vehicle RC:** Overall onboarding conversion is only **16.82%** (4,206 approved / 25,000 signups). The single largest volume leak is **Registration Certificate (RC) Stage 3**, where **6,102 captains (29.35% of all losses)** drop out—driven primarily by mechanical camera blur and OCR failures (63.63% of RC errors) on low-tier smartphones.
- **The Claimed 17.9% WhatsApp Nudge Lift is Confounded by Selection Bias:** The business claim that campaign CAMP_WA_002 drove a +17.90% lift is an artifact of immortal-time bias (100.0% of recipients had already survived DL and RC verification). When compared against stage-matched controls, the adjusted observational association is **+4.48% pts (95% CI: +3.12% to +5.84% pts)**, with 0% engagement lift among clickers. **Do not scale budget 5x without a controlled A/B test.**
- **Airport Shortage is a Nocturnal Deadhead Problem, Not an Acquisition Deficit:** The airport marketplace is 96.90% fulfilled during the day, but crashes to **47.20% fulfillment between 21:00 and 03:00** (accounting for 77.51% of unmet demand / 42,673 lost trips/mo). For suburban drops (SUB-07, SUB-11), return-fare access falls to **11.06% at night**, driving a **24.50% cancellation rate**.
- **Core Strategic Directive: Do NOT fund broad airport driver acquisition.** The airport is a leaky bucket exporting drivers into suburbs. Instead, deploy **Direct VAHAN API Auto-Fetch** for RC onboarding, **Run a Controlled RCT on WhatsApp Nudges**, and **Deploy Forward-Dispatch & Suburban Return Subsidies**.

1. ONBOARDING HEALTH & FUNNEL LEAK DIAGNOSIS

- **Funnel Conversion Reality (A2O & R2A):** Out of 25,000 signups, 20,789 clear DL (83.16%), 14,687 clear RC (58.75%), 10,854 clear Aadhaar (43.42%), 7,921 pass Permit/Doc-4 (31.68%), 5,612 clear Fitness (22.45%), 4,206 clear Insurance (16.82%), and **4,206 reach Final Approval (16.82% A2O)**. Post-approval activation (R2A) is **85.00%** (3,575 active captains). Among mature cohorts (> 15 days before extraction), conversion is **17.57%**; all 1,297 in-progress signups are isolated to the final 15 days.
- **The Biggest Volume Loss: Stage 3 (Registration Certificate — RC)** loses **6,102 captains (29.35% of all drop-offs)**, making it the primary system constraint.
- **The Most Important Fixable Leak:** 34.82% of captains fail RC verification at least once, and **50.10% abandon immediately after failure**. Root causes are heavily technical: **63.63% of RC errors** stem from blurry photos (39.54%) and OCR name/chassis mismatches (24.09%). This disproportionately hurts **low-tier phone users**, who convert at only **13.42%** (vs 21.81% on high-tier) and suffer 75.6% of blur failures.

2. CAMPAIGN EVALUATION: CAMP_WA_002

- **The Reality Behind the Claim:** CAMP_WA_002 sent WhatsApp messages to 8,673 captains. While recipients show an unadjusted 28.51% approval rate vs 10.61% for all non-recipients (a naive "+17.90% pt win"), **100.0% of treated captains had ALREADY passed DL and RC** before receiving the message.
- **Stage-Matched Observational Lift:** Comparing treated captains against untreated controls who also cleared DL+RC yields an approval rate of **28.51% vs 24.14% ($\Delta = +4.37\%$ pts)**. Adjusting for city, vehicle, channel, device tier, age, and month via logistic regression yields **+4.48% pts (95% CI: +3.12% to +5.84% pts, $p < 0.001$)**.
- **Causal Status & Confidence:** This is strictly an **observational association, not causal proof**. Captains who clicked the link converted at **28.24%**, identical to non-clickers (**28.70%**), proving the message acted as passive awareness rather than active friction removal.
- **Decision on 5x Budget Scaling: REJECT 5x budget scale.** Do not commit capital to linear broadcast blasts. Instead, execute a 50/50 randomized holdout experiment (N=3,305/arm) to test an automated 2-way failure-recovery bot.

3. AIRPORT SUPPLY & MARKETPLACE DYNAMICS

- **When and Where Shortage Occurs:** Undersupply is strictly nocturnal. Daytime (04:00–20:00) is healthy (**96.90% fulfillment**, ETA 3.8 min, Surge 1.05x). The **21:00–03:00 window collapses to 47.20% fulfillment** (26.24% at 23:00), with surge averaging 2.17x and ETAs hitting 10.0 minutes.
- **Magnitude of Deficit:** **42,673 requests go unfulfilled during the night window (77.51% of all unmet airport demand)**, averaging ~710 lost requests per night.
- **Post-Trip Trajectory & The Deadhead Trap:** Airport trips average 17.9 km (₹279 fare). However, **41.07% of trips drop passengers in suburban zones (SUB-07, SUB-11)**, where the night return-fare rate is only **11.06% (88.94% stranded rate)**. Suburban night cancellations spike to **24.50%** as drivers reject unpaid 30 km return deadheading.
- **Targeted Acquisition Decision: NOT RECOMMENDED.** Acquiring drivers around the airport creates a leaky bucket—drivers take suburban trips and are immediately drained from the catchment. We must fix return economics and turnaround speed first.

4. THREE RANKED STRATEGIC RECOMMENDATIONS

RANK	STRATEGIC ACTION	TARGET PROBLEM	EXPECTED IMPACT (SCENARIO)	COST & RISK	PRIMARY METRIC
#1	Direct VAHAN API Auto-Fetch & Blur Guidance Replace manual RC photo upload with instant RTO vehicle lookup via Reg Number + OTP; add client-side blur detection.	RC Stage 3 drop-off (6,102 drops; 63.6% blur/OCR failure share).	+75 to +85 Approved Captains / month (+450 to +510 over 6 mo). Formula: 3,465 DL passers × 16.82% Δpass × 28.64% downstream × 50% capture.	API fee (₹5–8/call); RTO downtime mitigated via manual fallback.	RC First-Time Pass Rate (≥ 82%, from 65.2%) & Low-Tier Pass (≥ 75%).
#2	Controlled RCT Experiment on Nudges Before 5x Scaling Halt broadcast blast; launch 50/50 RCT (N=3,305/arm) testing an event-driven 2-way WhatsApp failure-resolution bot.	Immortal-time bias in naive claim (+17.9% pts); true lift +4.5% pts; 0% clicker lift.	Protects ₹15–20L marketing budget ; sizes true incremental ROI; test 2-way bot.	Opportunity cost of holding out 3,305 captains during 6-week trial.	14-Day Incremental Approval Lift (≥ +3.0% pts, RCT verified; p < 0.05).
#3	Airport Forward-Dispatch & Suburban Return Subsidies Pre-match inbound airport drops 5–7 min before terminal arrival; provide ₹120 return allowance for SUB-07/SUB-11 night drops.	77.5% nocturnal airport shortage & suburban deadhead trap (88.9% night stranded rate).	+15,500 to +20,000 Fulfilled Trips / month (Forward-dispatch: +9k–12k; Return allowance: +6.5k–8k).	₹120 allowance funded via existing 2.17x night surge pool (net subsidy ≤ ₹45/trip).	Airport Night Fulfillment (≥ 70%, from 47.2%) & Suburb Cancels (< 10%).

5. ASSUMPTIONS & WHAT WOULD CHANGE MY ANSWER

- **Downstream Conversion Multiplier (28.64%):** Assumes captains rescued at RC pass downstream stages (Aadhaar → Insurance) at historical rates. *What changes answer:* If RC-failing drivers possess uninsurable vehicles (downstream pass < 15%), net approvals drop to +44/mo, shifting priority to Insurance FinTech financing.
- **Suburban Deadheading Elasticity:** Assumes night cancellations (24.50%) are driven by return economics (11.06% return fare rate). *What changes answer:* If cancellations stem from personal driver night-shift curfews rather than fuel economics, return subsidies will have lower elasticity.
- **Government VAHAN API Uptime:** Assumes ≥90% digital coverage across RTO databases in target cities. *What changes answer:* If rural vehicle registration records have high latency, on-device native camera guides become the primary path.